IGCSE 0607 Practice Questions — E6.4 — Combined Questions & Markscheme

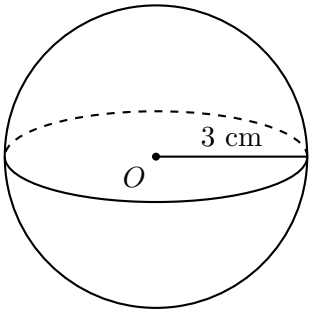
Surface area and volume

Total Questions: 18 — **Total Marks:** 138 Marks

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- **E6.4** Surface area and volume
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(11 Qs)*

1 The diagram shows a solid sphere with center O and radius 3 cm.



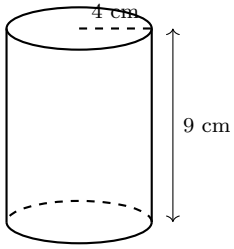
(a) Calculate the exact volume of the sphere.
Give your answer in terms of π.

..... cm^3 **[2]**

(b) Calculate the exact surface area of the sphere.
Give your answer in terms of π.

..... cm^2 **[2]**

2 The diagram shows a solid cylinder of base radius 4 cm and height 9 cm.



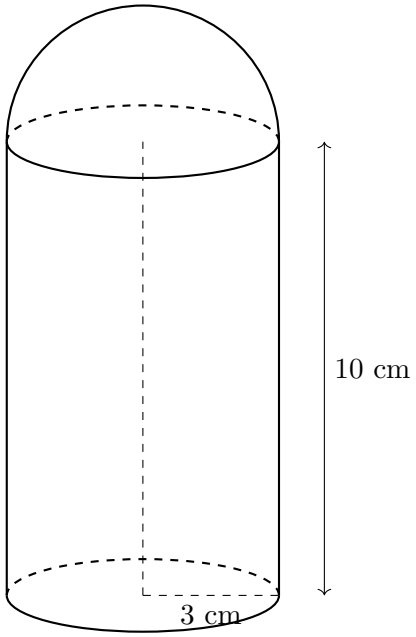
(a) Calculate the volume of the cylinder.
Give your answer in terms of π.

Volume = ... cm^3 **[2]**

(b) Calculate the total surface area of the solid cylinder.
Give your answer in terms of π.

Surface area = ... cm^2 **[3]**

3 A solid is made by joining a solid hemisphere of radius 3 cm to a solid cylinder of radius 3 cm and height 10 cm.



NOT TO SCALE

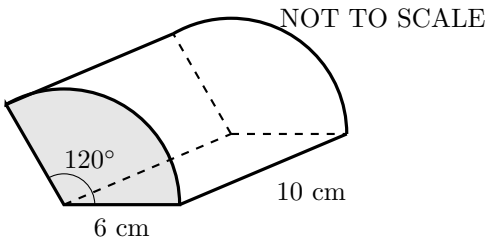
(a) Work out the volume of the solid.
Give your answer in terms of π.

..... cm^3 **[3]**

(b) Work out the total surface area of the solid.
Give your answer in terms of π.

..... cm^2 **[3]**

4 The diagram shows a solid prism of length 10 cm. The uniform cross-section of the prism is a sector of a circle of radius 6 cm and sector angle 120°.



(a) Giving your answers in terms of π, calculate

(i) the area of the sector cross-section,

..... cm^2 **[2]**

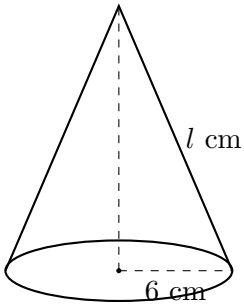
(ii) the volume of the prism.

..... cm^3 **[1]**

(b) Calculate the total surface area of the prism, giving your answer in the form $a + b\pi$, where a and b are integers.

..... cm^2 **[4]**

5 The diagram shows a solid cone.
The base radius of the cone is 6 cm and the slant height is l cm.
The total surface area of the cone is 96π cm^2.



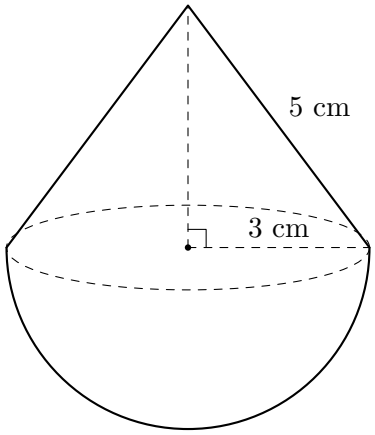
(a) Show that the slant height, l, is 10 cm.

[3]

(b) Work out the volume of the cone.
Give your answer in terms of π.

Volume = ... cm^3 **[4]**

6 The diagram shows a solid formed by joining a cone to a hemisphere with the same radius.
The radius of the hemisphere is 3 cm and the slant height of the cone is 5 cm.



(a) Calculate the exact volume of the solid.
Give your answer in terms of π.

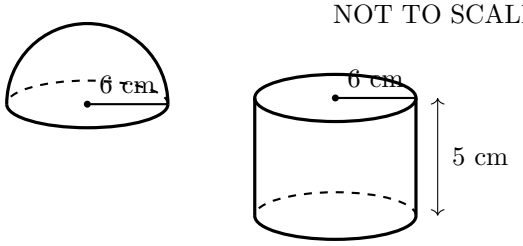
..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.
Give your answer in terms of π.

..... cm^2 **[3]**

7 The diagram shows a solid metal hemisphere of radius 6 cm and a solid metal cylinder of base radius 6 cm and height 5 cm.

NOT TO SCALE



(a) Giving your answers in terms of π, calculate

(i) the volume of the solid hemisphere,

..... cm^3 **[2]**

(ii) the total volume of metal in the hemisphere and the cylinder.

..... cm^3 **[2]**

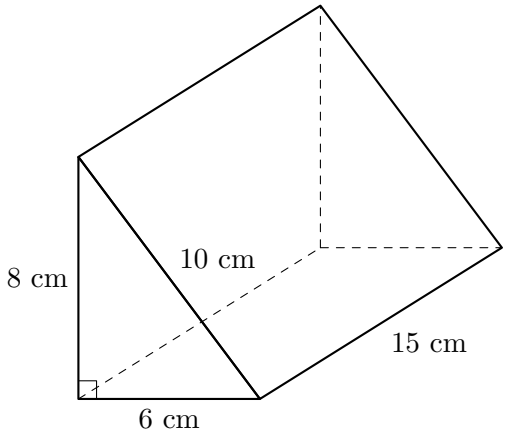
(b) The hemisphere and the cylinder are melted down and recast, with no loss of metal, into a single solid cone of base radius 9 cm. Show that the vertical height of the cone is 12 cm.

[2]

(c) Calculate the total surface area of the solid cone, giving your answer in terms of π.

..... cm^2 **[3]**

8 The diagram shows a solid triangular prism.
The cross-section of the prism is a right-angled triangle with sides 6 cm, 8 cm and 10 cm.
The length of the prism is 15 cm.



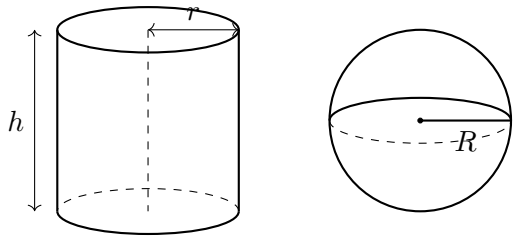
(a) Calculate the volume of the prism.

..... cm^3 **[2]**

(b) Calculate the total surface area of the prism.

..... cm^2 **[3]**

9 The diagram shows a solid cylinder and a solid sphere.



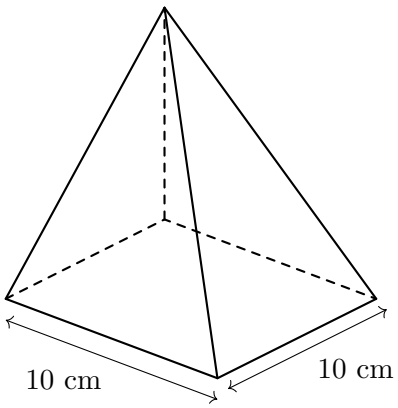
(a) The cylinder has radius $r = 5$ cm and height $h = 12$ cm.
Work out the total surface area of the cylinder.

Surface area = ... cm^2 **[3]**

(b) The sphere has the same volume as the cylinder.
Calculate the radius, R, of the sphere.

$R =$ ..... cm **[3]**

10 The diagram shows a solid pyramid with a square base.
The base has sides of length 10 cm.
The volume of the pyramid is 400 cm^3.



NOT TO SCALE

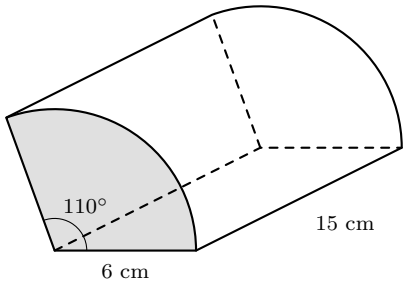
(a) Calculate the perpendicular height of the pyramid.

..... cm **[2]**

(b) Calculate the total surface area of the pyramid.

..... cm^2 **[4]**

11 The diagram shows a prism of length 15 cm.
The cross-section of the prism is a sector of a circle with radius 6 cm and sector angle 110°.



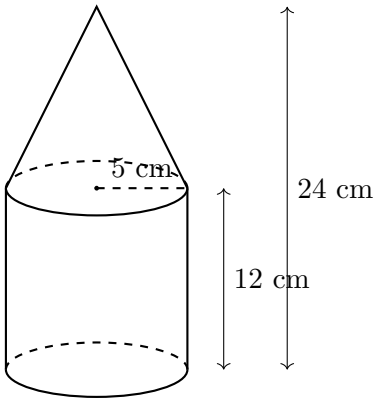
(a) Calculate the volume of the prism.

Volume = ... cm^3 **[3]**

(b) Calculate the total surface area of the prism.

Surface area = ... cm^2 **[3]**

12 A solid is formed by attaching a right circular cone to the top of a solid cylinder.
The cylinder and the base of the cone both have a radius of 5 cm.
The height of the cylinder is 12 cm and the total height of the solid is 24 cm.



(a) Calculate the volume of the solid.

..... cm^3 **[4]**

(b) Calculate the total surface area of the solid.

..... cm^2 **[5]**

13 A solid metal cylinder has radius x cm and height $3x$ cm.
The cylinder is melted down and completely recast into solid spheres.
Each sphere has a radius of $\frac{x}{2}$ cm.

(a) Find an expression, in terms of x and π, for the volume of the cylinder.
Give your answer in its simplest form.

..... cm^3 **[2]**

(b) Find an expression, in terms of x and π, for the volume of one sphere.
Give your answer in its simplest form.

..... cm^3 **[2]**

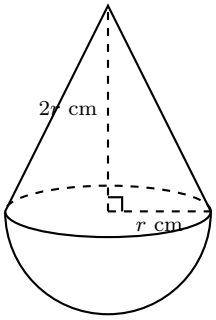
(c) Work out the number of complete spheres that can be made.

..... **[1]**

(d) The total surface area of the original cylinder is 140 cm^2.
Work out the value of x.

$x =$ ..... **[3]**

14 A solid toy is made by joining a solid right circular cone to the flat circular face of a solid hemisphere.
Both the cone and the hemisphere have radius r cm.
The perpendicular height of the cone is $2r$ cm.



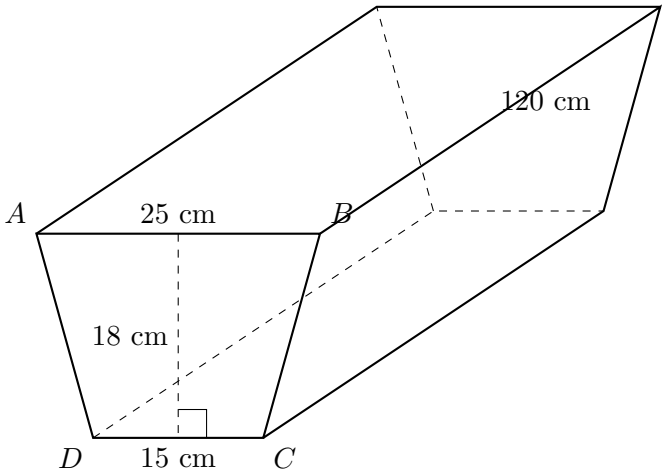
(a) The total volume of the solid toy is 450π cm^3.
Show that $r = 3\sqrt[3]{12.5}$.

[4]

(b) Calculate the total surface area of the solid toy.
Give your answer correct to 3 significant figures.

Surface area = ... cm^2 **[4]**

15 The diagram shows a water trough in the shape of a prism of length 120 cm.
The cross-section of the prism is a trapezium $ABCD$.
AB is parallel to DC.
$AB = 25$ cm, $DC = 15$ cm and the perpendicular distance between AB and DC is 18 cm.



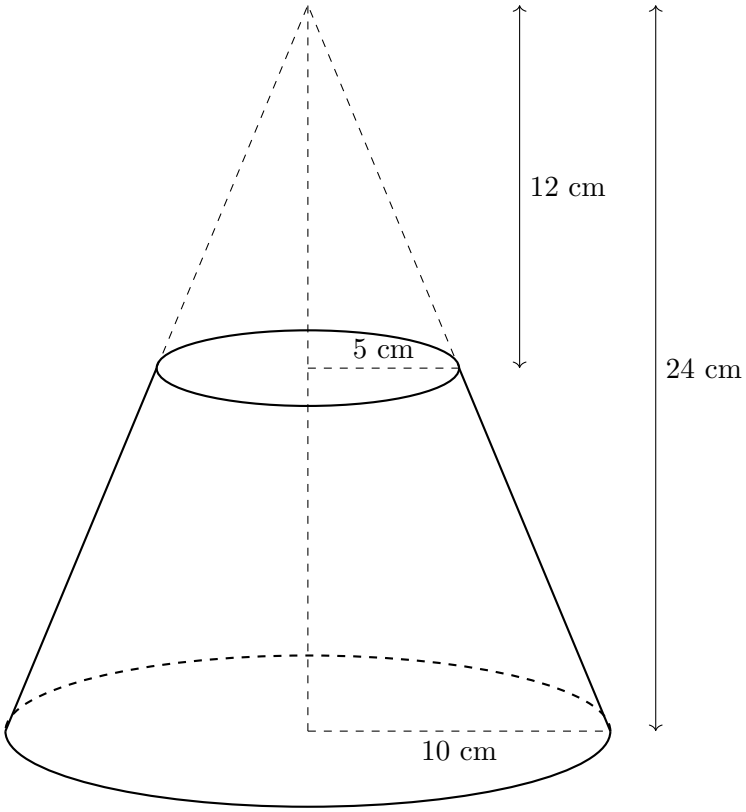
(a) Calculate the volume of the trough in litres.

Volume = ... litres **[4]**

(b) The trough is initially completely empty.
Water flows into the trough at a constant rate of 0.8 litres per second.
Calculate the depth of the water in the trough after 30 seconds.

Depth = ... cm **[5]**

16 A solid is made by removing a small cone from a large cone to form a frustum.
The large cone has a base radius of 10 cm and a perpendicular height of 24 cm.
The small cone has a base radius of 5 cm and a perpendicular height of 12 cm.



NOT TO SCALE

(a) Calculate the volume of the frustum.

..... cm^3 **[3]**

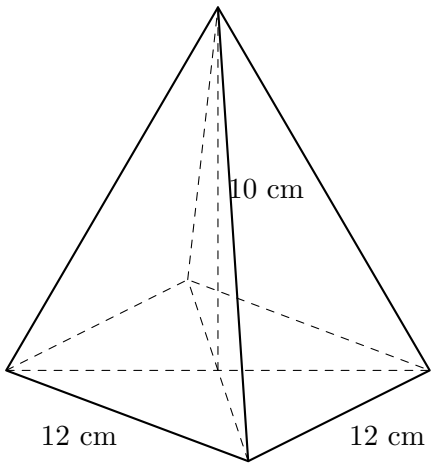
(b) Calculate the total surface area of the solid frustum.

..... cm^2 **[4]**

(c) The solid frustum is melted down to form solid spheres of radius 3 cm.
Calculate the maximum number of complete spheres that can be made.

..... **[3]**

17 A solid metal pyramid has a square base of side 12 cm and a perpendicular height of 10 cm.



(a) Calculate the volume of the pyramid.

..... cm^3 **[2]**

(b) Calculate the total surface area of the pyramid.

..... cm^2 **[4]**

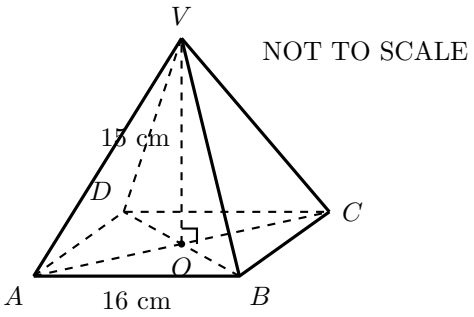
(c) The metal pyramid is melted down and recast to make solid spheres of radius 2.5 cm.
Calculate the number of complete spheres that can be made.

..... **[3]**

(d) A mathematically similar pyramid has a volume of 1620 cm^3.
Calculate the perpendicular height of this similar pyramid.

..... cm **[3]**

18 The diagram shows a solid right pyramid $VABCD$ with a horizontal square base $ABCD$ of side 16 cm.
O is the centre of the base and the vertical height $VO = 15$ cm.



(a) Calculate the volume of pyramid $VABCD$.

..... cm^3 **[2]**

(b) Calculate the total surface area of pyramid $VABCD$.

..... cm^2 **[3]**

(c) The pyramid is cut by a horizontal plane parallel to the base $ABCD$ at a vertical height of 9 cm above the base.
The top small pyramid is removed, leaving a solid frustum.

(i) Calculate the volume of the frustum.

..... cm^3 **[3]**

(ii) Calculate the total surface area of the frustum.

..... cm^2 **[3]**

(d) Calculate the angle that the slant edge VA makes with the horizontal base $ABCD$, giving your answer correct to 1 decimal place.

..... $^\circ$ **[3]**

1 **(a)** $\frac{4}{3} \times \pi \times 3^3$ **[M1]**

36π **[A1]**

(b) $4 \times \pi \times 3^2$ **[M1]**

36π **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $\pi \times 4^2 \times 9$ **[M1]**

144π **[A1]**

(b) Curved surface area $= 2 \times \pi \times 4 \times 9 = 72\pi$ **[M1]**

Area of two circular ends $= 2 \times \pi \times 4^2 = 32\pi$ **[M1]**

104π **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** Volume of cylinder $= \pi \times 3^2 \times 10$ $(= 90\pi)$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 3^3$ $(= 18\pi)$ **[M1]**

108π **[A1]**

(b) Curved surface area of cylinder $= 2 \times \pi \times 3 \times 10$ $(= 60\pi)$ **[M1]**

Area of circular base $= \pi \times 3^2$ $(= 9\pi)$ AND Curved surface area of hemisphere $= 2 \times \pi \times 3^2$ $(= 18\pi)$ **[M1]**

87π **[A1]**

[Back to Question] *[Total Marks: 6]*

4 **(a)** **(i)** $\frac{120}{360} \times \pi \times 6^2$ **[M1]**

12π **[A1]**

(ii) $12\pi \times 10 = 120\pi$ **[B1]**

(b) Curved arc length $= \frac{120}{360} \times 2\pi(6) = 4\pi$ cm **[M1]**

Curved surface area $= 4\pi \times 10 = 40\pi\ cm^2$ **[M1]**

Area of two rectangular faces $= 2 \times (6 \times 10) = 120\ cm^2$ **[B1]**

Total surface area $= 120 + 40\pi + 2(12\pi) = 120 + 64\pi$ **[A1]**

[Back to Question] *[Total Marks: 7]*

5 **(a)** Area of base $= \pi \times 6^2 = 36\pi$ **[M1]**

Curved surface area $= \pi \times 6 \times l = 6\pi l$ **[M1]**

$36\pi + 6\pi l = 96\pi$ leading to $6l = 60$ so $l = 10$ **[A1]**

(b) $h^2 + 6^2 = 10^2$ **[M1]**

$h = \sqrt{100 - 36} = \sqrt{64} = 8$ **[A1]**

Volume $= \frac{1}{3} \times \pi \times 6^2 \times 8$ **[M1]**

96π **[A1]**

[Back to Question] *[Total Marks: 7]*

6 (a) Use of Pythagoras to find the height of the cone: $\sqrt{5^2 - 3^2}$ $(= 4)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 3^2 \times 4$ $(= 12\pi)$ **[M1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3} \times \pi \times 3^3$ $(= 18\pi)$ **[M1]**

30π **[A1]**

(b) Curved surface area of cone $= \pi \times 3 \times 5$ $(= 15\pi)$ **[M1]**

Surface area of hemisphere $= 2 \times \pi \times 3^2$ $(= 18\pi)$ **[M1]**

33π **[A1]**

[Back to Question] *[Total Marks: 7]*

7 (a) (i) $\frac{1}{2} \times \frac{4}{3}\pi \times 6^3$ **[M1]**

144π **[A1]**

(ii) Volume of cylinder $= \pi \times 6^2 \times 5 = 180\pi$ **[M1]**

Total volume $= 144\pi + 180\pi = 324\pi$ **[A1]**

(b) $\frac{1}{3}\pi \times 9^2 \times h = 324\pi$ giving $27\pi h = 324\pi$ **[M1]**

$h = \frac{324}{27} = 12$ cm **[A1]**

(c) Slant height $l = \sqrt{9^2 + 12^2} = 15$ cm **[M1]**

Total surface area $= \pi(9)(15) + \pi(9^2) = 135\pi + 81\pi$ **[M1]**

216π **[A1]**

[Back to Question] *[Total Marks: 9]*

8 (a) $\frac{1}{2} \times 6 \times 8 \times 15$ **[M1]**

360 **[A1]**

(b) Area of two triangular ends $= \frac{1}{2} \times 6 \times 8 \times 2$ $(= 48)$ **[M1]**

Area of three rectangular faces $= 15 \times (6 + 8 + 10)$ or $(15 \times 6) + (15 \times 8) + (15 \times 10)$ $(= 360)$ **[M1]**

408 **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** Curved surface area $= 2 \times \pi \times 5 \times 12$ **[M1]**

Total surface area $= 2 \times \pi \times 5 \times 12 + 2 \times \pi \times 5^2$ **[M1]**

534 or 534.07... or 170π **[A1]**

(b) Volume of cylinder $= \pi \times 5^2 \times 12 = 300\pi$ (942...) **[M1]**

$\frac{4}{3}\pi R^3 = 300\pi$ **[M1]**

$R = \sqrt[3]{225} = 6.08$ or 6.082... **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** $\frac{1}{3} \times 10^2 \times h = 400$ **[M1]**

12 **[A1]**

(b) Distance from centre of base to midpoint of side = 5 cm **[B1]**

Slant height of a triangular face $= \sqrt{12^2 + 5^2}$ $(= 13)$ **[M1]**

Area of one triangular face $= \frac{1}{2} \times 10 \times 13$ $(= 65)$ **[M1]**

360 **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Calculates cross-sectional sector area: $\frac{110}{360} \times \pi \times 6^2 = 11\pi$ $(= 34.557...)$ **[M1]**

Multiplies by length: $11\pi \times 15$ **[M1]**

518 (accept 518.3 to 518.4) **[A1]**

(b) Calculates curved arc length: $\frac{110}{360} \times 2 \times \pi \times 6 = \frac{11\pi}{3}$ $(= 11.519...)$ **[M1]**

Sums all five faces: $2(11\pi) + 15 \times \left(\frac{11\pi}{3} + 6 + 6\right)$ **[M1]**

422 (accept 421.9 to 422.0) **[A1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** Height of cone $= 24 - 12 = 12$ cm **[B1]**

Volume of cylinder $= \pi \times 5^2 \times 12$ $(=300\pi$ or $942.4\ldots)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ $(=100\pi$ or $314.1\ldots)$ **[M1]**

1260 (or 1256 to 1257 or 400π) **[A1]**

(b) Slant height of cone $(l) = \sqrt{5^2 + 12^2} = 13$ **[M1]**

Curved surface area of cone $= \pi \times 5 \times 13$ $(=65\pi$ or $204.2\ldots)$ **[M1]**

Curved surface area of cylinder $= 2 \times \pi \times 5 \times 12$ ($=120\pi$ or $376.9\ldots$) **[M1]**

Area of circular base $= \pi \times 5^2$ ($=25\pi$ or $78.5\ldots$) **[M1]**

660 (or $659.7\ldots$ or 210π) **[A1]**

[Back to Question] *[Total Marks: 9]*

13 (a) $\pi \times (x)^2 \times 3x$ **[M1]**

$3\pi x^3$ **[A1]**

(b) $\frac{4}{3} \times \pi \times \left(\frac{x}{2}\right)^3$ **[M1]**

$\frac{\pi x^3}{6}$ **[A1]**

(c) 18 **[B1]**

(d) Total surface area $= 2\pi x^2 + 2\pi x(3x)$ **[M1]**

$8\pi x^2 = 140$ **[M1]**

$x = 2.36$ (or $2.360\ldots$) **[A1]**

[Back to Question] *[Total Marks: 8]*

14 (a) Volume of cone $= \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$ **[B1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi r^3$ **[B1]**

Equates total volume: $\frac{2}{3}\pi r^3 + \frac{2}{3}\pi r^3 = 450\pi$ leading to $\frac{4}{3}r^3 = 450$ **[M1]**

Solves for r: $r^3 = 337.5 = 27 \times 12.5$, therefore $r = 3\sqrt[3]{12.5}$ **[A1]**

(b) Finds slant height of cone: $l = \sqrt{r^2 + (2r)^2} = r\sqrt{5}$ **[M1]**

Sets up total surface area expression: $\pi rl + 2\pi r^2 = \pi r^2(\sqrt{5} + 2)$ **[M1]**

Evaluates $r = 3\sqrt[3]{12.5} = 6.96238...$ and substitutes into area formula **[M1]**

645 (accept 645.0 to 645.4) **[A1]**

[Back to Question] *[Total Marks: 8]*

15 (a) Area of trapezium $= \frac{1}{2} \times (15 + 25) \times 18$ **[M1]**

360 **[A1]**

Volume $= 360 \times 120 = 43200$ cm^3 **[M1]**

43.2 **[A1]**

(b) Volume of water after 30 seconds $= 0.8 \times 30 = 24$ litres $= 24000$ cm^3 **[M1]**

Let depth be h and the width of water surface be w.

$w = 15 + \frac{10}{18}h = 15 + \frac{5}{9}h$ **[M2]**

Area of cross-section of water $= \frac{1}{2} \times (15 + 15 + \frac{5}{9}h) \times h = 15h + \frac{5}{18}h^2$ **[M1]**

Volume $= (15h + \frac{5}{18}h^2) \times 120 = 24000$

$1800h + \frac{100}{3}h^2 = 24000$

$h^2 + 54h - 720 = 0$ **[M1]**

$(h - 11.23...)(h + 65.23...) = 0$

$h = 11.2$ or $11.23...$ **[A1]**

[Back to Question] *[Total Marks: 9]*

16 **(a)** Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24$ $(= 800\pi)$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ $(= 100\pi)$ **[M1]**

2200 or 2199.1 to 2199.2 **[A1]**

(b) Slant height of large cone $= \sqrt{10^2 + 24^2}$ $(= 26)$ AND Slant height of small cone $= \sqrt{5^2 + 12^2}$ $(= 13)$ **[M1]**

Curved surface area of frustum $= \pi \times 10 \times 26 - \pi \times 5 \times 13$ $(= 195\pi)$ **[M1]**

Total surface area $= 195\pi + \pi \times 10^2 + \pi \times 5^2$ **[M1]**

1010 or 1005.3 . . . **[A1]**

(c) Volume of one sphere $= \frac{4}{3} \times \pi \times 3^3$ $(= 36\pi$ or $113.1...)$ **[M1]**

$\frac{700\pi}{36\pi}$ or $\frac{2199.1}{113.1}$ **[M1]**

19 **[A1]**

[Back to Question] *[Total Marks: 10]*

17 **(a)** $\frac{1}{3} \times 12^2 \times 10$ **[M1]**

480 **[A1]**

(b) Use of Pythagoras to find slant height: $\sqrt{10^2 + 6^2}$ **[M1]**

$\sqrt{136}$ or 11.6... or 11.7 **[A1]**

Area of four triangular faces $= 4 \times \frac{1}{2} \times 12 \times$ their $\sqrt{136}$ $(= 279.8...)$ **[M1]**

424 or 423.8... to 423.9 **[A1]**

(c) Volume of one sphere $= \frac{4}{3} \times \pi \times 2.5^3$ $(= 65.4...)$ **[M1]**

$\frac{480}{\text{their } 65.4...}$ **[M1]**

7 **[A1]**

(d) $\frac{1620}{480}$ or 3.375 **[M1]**

$\sqrt[3]{3.375}$ or 1.5 **[M1]**

15 **[A1]**

[Back to Question] *[Total Marks: 12]*

18 **(a)** $\frac{1}{3} \times 16^2 \times 15$ **[M1]**

1280 **[A1]**

(b) Slant height of triangular face $= \sqrt{15^2 + 8^2} = 17$ cm **[M1]**

Total surface area $= 16^2 + 4 \times \left(\frac{1}{2} \times 16 \times 17\right) = 256 + 544$ **[M1]**

800 **[A1]**

(c) **(i)** Height of removed pyramid $= 15 - 9 = 6$ cm, giving linear scale factor $= \frac{6}{15} = 0.4$ **[M1]**

Volume of frustum $= 1280 - 0.4^3 \times 1280 = 1280 - 81.92$ **[M1]**

1198.08 (or 1200) **[A1]**

(ii) Top square side $= 0.4 \times 16 = 6.4$ cm and trapezoidal slant height $= 0.6 \times 17 = 10.2$ cm **[M1]**

Total surface area $= 16^2 + 6.4^2 + 4 \times \frac{1}{2}(16 + 6.4) \times 10.2 = 256 + 40.96 + 456.96$ **[M1]**

753.92 (or 754) **[A1]**

(d) $OA = \sqrt{8^2 + 8^2} = \sqrt{128}$ $(= 11.313\ldots)$ cm **[M1]**

$\tan \angle VAO = \frac{15}{\sqrt{128}}$ **[M1]**

53.0 **[A1]**

[Back to Question] *[Total Marks: 14]*